

FINAL PROJECT: Pharmacy Claims



Ronit Hara Yandra

College of Professional Studies, Northeastern University

ALY 6030: Data Warehousing and SQL

Prof. Kayal Chadrsekaran

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Part 1 – Normalization

For normalizing the dataset into 1NF format, first I have merged the columns fill_date1, fill_date2 and fill_date3 into fill_date similarly I have merged copay and insurancepaid columns as well.

	A	B	C	D	E	F	G
1	member_id	drug_ndc	drug_form_code	drug_brand_generic_code	fill_date	copay	insurancepaid
2	10001	433530848	TB		1 10/31/17	15	50
3	10001	433530848	TB		1 2/22/18	15	48
4	10001	433530848	TB		1 5/8/18	15	55
5	10002	545695193	OS		1 6/14/18	50	130
6	10003	545693828	TB		2 12/30/17	35	250
7	10003	545693828	TB		2 5/16/18	35	322
8	10004	185085302	TC		1 11/9/17	15	600
9	10004	185085302	TC		1 12/8/17	15	712
10	10001	545693828	TB		2 1/15/18	20	650
11	10001	545693828	TB		2 2/14/18	20	648
12	10001	545693828	TB		2 3/13/18	20	648

To normalize it into 2NF I have formed a different csv table for patients naming it into dimension_member csv;

member_id	member_first_name	member_last_name	member_birth_date	member_age	member_gender
10001	David	Dennison	14-06-1946	72	M
10002	John	Smith	02-01-1962	56	M
10003	Jane	Doe	04-05-1982	36	F
10004	Elaine	Rogers	12-10-1983	34	F

And drug table;

drug_ndc	drug_name	drug_form_code	drug_form_desc	drug_brand_generic_code	drug_brand_generic_desc
433530848	Risperidone	TB	Tablet	1	Generic
545695193	Amoxicillin	OS	Oral Solution	1	Generic
545693828	Ambien	TB	Tablet	2	Brand
185085302	Diprosone	TC	Topical Cream	1	Generic

Further for 3NF format I split the above table for drug_ndc, drug_brand_generic and for drug form. Below I have showcased my normalized tables in 3NF format.

Dimension_drug csv

	A	B
1	drug_ndc	drug_name
2	185085302	Diprosone
3	433530848	Risperidone
4	545693828	Ambien
5	545695193	Amoxicillin

Dimension_drugadministration csv;

	A	B
1	drug_form_code	drug_form_desc
2	TB	Tablet
3	OS	Oral Solution
4	TC	Topical Cream

Dimension_drugbrandgeneric csv;

	A	B
1	drug_brand_generic_code	drug_brand_generic_desc
2		1 Generic
3		2 Brand

In my fact_patientsprescription table:

- the fact variable copay is a semi-additive fact variable and,
- insurancepaid variable is an additive fact variable.

In my fact table each fact row represents the payments done by the patient as well as through health insurance coverage on the specific fill dates.

Part 2 – Primary and Foreign Key setup

```

/*
setting up the primary and foreign keys for the below tables
Also setting up the constraints to follow the referential integration like setting up
on delete cascade or on update cascade
*/

alter table dimension_drug
add primary key (drug_ndc);

-- Modify the data type of drug_form_code
alter table dimension_drugadministration
modify column drug_form_code varchar(100);

alter table dimension_drugadministration
add primary key (drug_form_code);

alter table dimension_drugbrandgeneric
add primary key (drug_brand_generic_code);

alter table dimension_member
add primary key (member_id);

24
25 -- Add primary key
26 • alter table fact_patientsprescription
27 modify column drug_form_code varchar(100);
28

```

```

28
29 • alter table fact_patientsprescription
30   add primary key (member_id, drug_ndc, drug_form_code, drug_brand_generic_code);
31
32 -- Add foreign key constraints
33 • alter table fact_patientsprescription
34   add constraint FK_member
35   foreign key (member_id) references dimension_member (member_id)
36   on delete cascade on update cascade;
37
38 • alter table fact_patientsprescription
39   add constraint FK_drug
40   foreign key (drug_ndc) references dimension_drug (drug_ndc)
41   on delete cascade on update cascade;
42
43 • alter table fact_patientsprescription
44   add constraint FK_drugadministration
45   foreign key (drug_form_code) references dimension_drugadministration (drug_form_code)
46   on delete cascade on update cascade;
47
48 • alter table fact_patientsprescription
49   add constraint FK_drugbrandgeneric
50   foreign key (drug_brand_generic_code) references dimension_drugbrandgeneric (drug_brand_generic_code)
51   on delete cascade on update cascade;
52

```

a) There were totally 5 tables made from the original after normalization. The primary keys for each table are as follows:

- **Dimension_member**; this table has a natural primary key which is the member_id holding a unique id for the members who prescribed for a drug consisting of their details.
- **Dimension_drug**; it also consists of a natural primary key called drug_ndc which is a unique drug identification code for each drug.
- **Dimension_drugadministration**; this table consists of a natural primary key called drug_form_code which holds the detail related to the drug administration, whether the administered drug is a tablet, topical cream, or oral solution.
- **Dimension_drugbrandgeneric**; drug_brand_generic_code is the natural primary key here which stores the information which can be linked to the fact table to indicate whether the prescribed drug is brand name or generic equivalent.
- **Fact_patientsprescription**; fill_date would be the surrogate key and all the above four tables are linked to this through a foreign key.

b) The foreign keys were assigned to the fact table which acts as the main table for the database, the reason being that table holds the important detail of the fill dates of the member filling the medication, the copay paid by the patient on that fill date, and insurance paid through the patient's pharmacy coverage plan.

The foreign keys for the fact_patientsprescription table are:

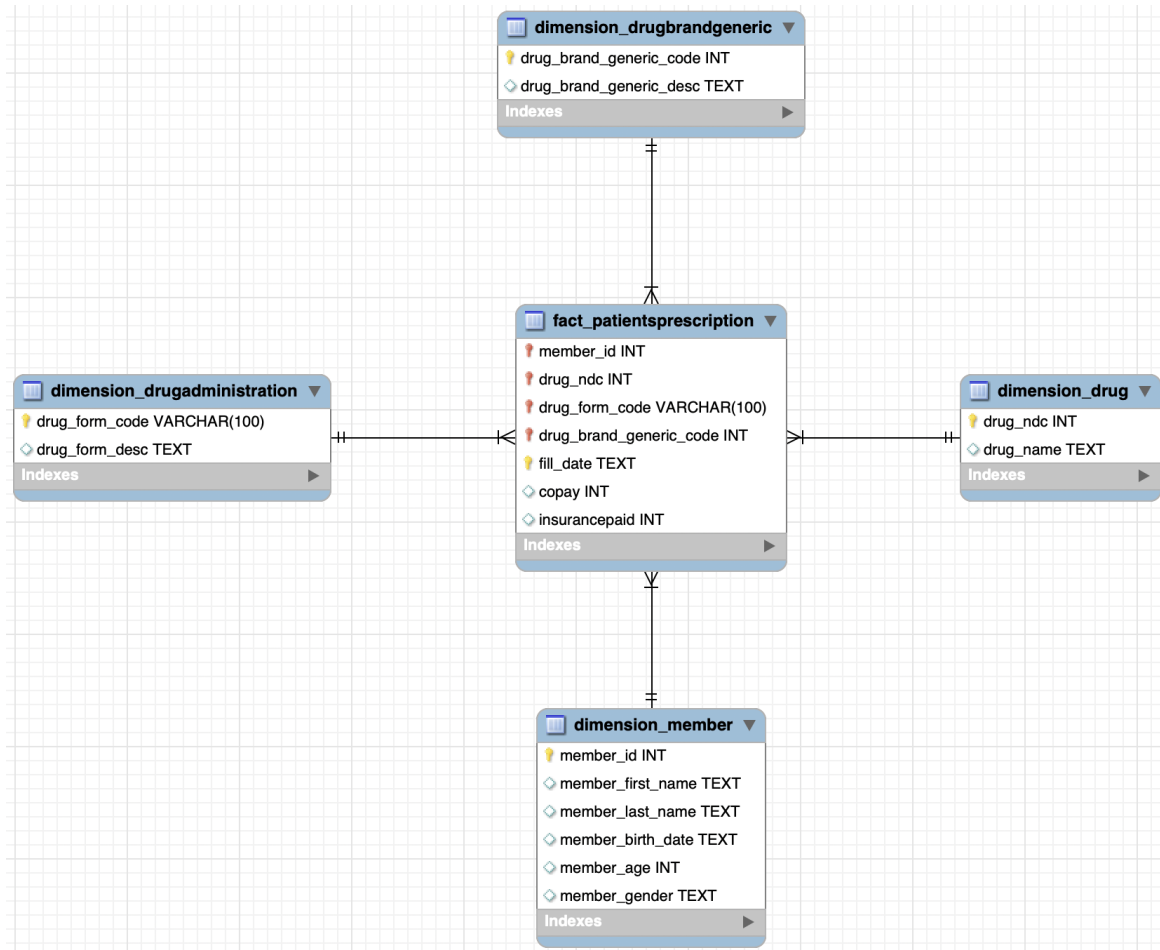
- 1) member_id,
- 2) drug_ndc,
- 3) drug_form_code,
- 4) drug_brand_generic_code

c) For the above keys it is better to set them up as cascade upon deletion or update, the reason being:

- To maintain the data integrity, when the data in any of the child tables are updated or deleted then these changes are reflected upon the main fact_patientsprescription table as well to maintain the data consistency.
- The other reason being to streamline data management, by setting up the cascade condition. To assure compliance with corporate rules and laws by enforcing cascade actions, it simplifies database operations like bulk updates and deletes by cascading changes to the fact_patientsprescription table without requiring individual treatment of each linked item.

PART 3 – Entity Relationship Diagram (ERD)

Down below is the ERD for the database in star schema format.



PART 4 – Analytics and Reporting

Query 1:

```

54      #query no. 1
55      * select d.drug_name, count(f.drug_ndc) as total_prescriptions
56      from fact_patientsprescription f
57      join dimension_drug d on f.drug_ndc = d.drug_ndc
58      group by d.drug_name
59      order by total_prescriptions desc;

```

Output:

	drug_name	total_prescriptions
▶	Ambien	5
	Risperidone	3
	Diprosone	2
	Amoxicillin	1

For the question asked in the tasks there were totally 5 prescriptions filled for the drug Ambience.

Query 2:

```

61      #query no. 2
62      * select
63      ⊖ case
64      | when m.member_age > 65 then "age 65+"
65      | else "< 65"
66      | end as age_group,
67      count(f.drug_ndc) total_prescription,
68      count(distinct f.member_id) total_members,
69      sum(f.copay) as sums_copay, sum(f.insurancepaid) as suminsurancepaid
70      from fact_patientsprescription f
71      join dimension_member m on f.member_id = m.member_id
72      group by age_group;

```

Output:

	age_group	total_prescription	total_members	sums_copay	suminsurancepaid
▶	< 65	5	3	150	2014
	age 65+	6	1	105	2099

From the above results for the query and further answering the questions asked in the task; there is only 1 person who is above 65 years old and they filled in total 6 prescriptions.

Query 3:

```

74  #query no. 3
75  •  select member_id, member_first_name, member_last_name, drug_name, fill_date as most_recent_fill_date,
76     insurancepaid as most_recent_insurance_paid
77  ⊖  from (
78     select f.member_id, m.member_first_name, m.member_last_name, d.drug_name, f.fill_date, f.insurancepaid,
79     row_number() over (partition by f.member_id order by str_to_date(fill_date, '%m/%d/%Y') desc) as rn
80     from fact_patientsprescription f
81     inner join dimension_member m on f.member_id = m.member_id
82     inner join dimension_drug d on f.drug_ndc = d.drug_ndc
83     ) as recent_entry
84     where rn =1;

```

Output:

member_id	member_first_name	member_last_name	drug_name	most_recent_fill_date	most_recent_insurance_paid
10001	David	Dennison	Risperidone	5/8/18	55
10002	John	Smith	Amoxicillin	6/14/18	130
10003	Jane	Doe	Ambien	5/16/18	322
10004	Elaine	Rogers	Diprosone	12/8/17	712

Through the above results obtained, for the member id of 10003 the most recent drug listed on their recent fill date is Ambien and their insurance paid around \$322 for the medication.

SUMMARY

In this report a proper dimensional modelling was performed starting with normalizing the dataset to its 3NF form by segregating the facts and dimension tables, then assigning whether to either cascade the entries or set null or restrict on update or delete based on the business queries specified and setting up the foreign keys and primary keys for each table. Few tasks were required to be performed in SQL query to answer the questions. All the results are incorporated above and for further assistance additional documents are attached.

REFERENCES:

Referential Integrity Constraint | Violation. Retrieved 22 June 2023,

<https://www.gatevidyalay.com/referential-integrity-constraint-violation/Links to an external site>

Ceccarini, G.D. (2019) What are additive, semi-additive and non-additive measures?, Medium.

Available at: <https://medium.com/@gdceccarini/what-are-additive-semi-additive-and-non-additive-measures-8c2eb50759f3> (Accessed: 13 June 2023)